

MANISH MAN SHRESTHA

Firmware & Embedded Systems Engineer | IoT Developer

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Hiring Team

LMK Recruiting Solutions - Irvine, CA

Re: Senior Embedded Systems & Controls Engineer (Irvine, CA)

Dear Hiring Manager,

I am excited to apply for the Senior Embedded Systems & Controls Engineer position. My background sits right at the intersection this role needs: 15 years of real-time embedded firmware, a research career in aerospace structural health monitoring, and hands-on control-systems and power-electronics work.

Here is how my experience maps to the role:

- **Real-time, deterministic embedded C/C++:** I build real-time firmware in Embedded C/C++ on ARM Cortex-M microcontrollers (including NXP), including motor-control firmware using PWM, timers, ADC feedback, and FreeRTOS scheduling under strict, deterministic timing constraints.
- **Control architecture and sequencing logic:** I have designed embedded control and state/sequencing logic for multi-subsystem coordination - from a multi-axis servo-controlled robotic arm to teleoperated motion control - and taught undergraduate Control Systems.
- **Aerospace structural health monitoring (health-monitoring frameworks):** At KAIST I developed embedded SHM firmware for aircraft structures on a dual-core ARM Cortex-M4 (LPC4370) with 6-channel acquisition at 5 MSPS and on-device FFT/IIR/FIR DSP for real-time diagnostics - directly aligned with health-monitoring and diagnostic frameworks.
- **MATLAB modeling/validation and power electronics:** I prototype and validate signal-processing and control algorithms in MATLAB before embedded integration, and I have built power-electronics firmware including an MPPT solar battery charger and servo/DC motor control.
- **Hardware bring-up and cross-functional debugging:** I lead board bring-up and integration over SPI, I2C, UART, and Ethernet, debugging across hardware, firmware, and software layers with oscilloscopes, logic analyzers, and JTAG/SWD.

I am glad to work on-site in Irvine, and I am eager to bring my aerospace health-monitoring and real-time control experience to your team. Thank you for your consideration.

Sincerely,

Manish Man Shrestha